

SHUBHAM KISHOR KALE

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AI/Data Engineer building agentic RAG/LLM systems and large-scale ELT pipelines across healthcare, finance, and insurance domains.

EDUCATION

Stony Brook University New York, USA	Aug 2024 - Dec 2025
<i>Master of Science in Data Science</i>	GPA: 3.78/4
NMIMS University Mumbai, India	Jul 2019 - May 2023
<i>Bachelor of Technology in Data Science</i>	CGPA: 3.56/4

TECHNICAL SKILLS

Languages: Python, Java, C/C++, R, SQL, JavaScript, TypeScript, Bash
Concepts & Domains: AI Engineering, Data Engineering, Data Science, Data Analysis, Cloud Computing, Machine Learning, Deep Learning, NLP, Generative AI (RAG, LLM Agents, MCP, Fine-tuning), Statistical Analysis, Causal Inference, A/B Testing, Time Series Analysis, Data Structures & Algorithms, System Design, REST APIs, Server-Sent Events (SSE), OOP & Design Patterns, ETL, Data Ingestion, Data Warehousing, Data Modeling, Data Architecture, Data Security, Distributed Systems, Finance, Healthcare
Frameworks, Tools & Cloud: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, Claude API, Gemini API, LangChain, LangGraph, Pandas, NumPy, Spark, Kafka, Docker, Kubernetes, Apache Airflow, Dagster, dbt, AWS (S3, Redshift, IAM), Azure, GCP (BigQuery), Snowflake, PostgreSQL, MongoDB, FastAPI, Flask, Streamlit, React, Next.js, Pydantic, pytest, Git, GitHub Actions (CI/CD), Tableau, Power BI

WORK EXPERIENCE

Quantegy Analytics New York, USA	May 2025 - Present
<i>Analytics Engineer</i>	

- Shipped an LLM-powered navigation assistant (Claude API) into a production analytics platform serving 50+ daily users across 5 markets, routing natural-language queries to the right dashboard via a structured page-catalog context.
- Delivered production Streamlit platforms for 2 clients: a healthcare platform that cut a 3-day Excel reporting cycle to same-day, and separately a correctional healthcare app with per-county group-based access control for county and sheriff's office users.
- Engineered Dagster-orchestrated ELT pipelines ingesting 4M+ patient and financial records from 15+ Redshift tables into Snowflake, powering analytics across 3 healthcare markets.
- Owned end-to-end data engineering for a correctional healthcare client: Dagster pipelines ingesting 120+ tables of vendor EHR and CSV data into structured Snowflake schemas across 15 counties.
- Resolved an extraction bottleneck via parallel multi-table ingestion (~5x throughput) with REST API integration, S3 archival, and Pydantic schema validation in a Dockerized CLI framework adopted by 4+ teams.
- Productionized scheduling on Airflow with full pytest/httpx coverage validating 10M+ records across 10+ tables.
- Secured 2 client environments end to end: AWS IAM/VPN access policies, Secrets Manager credentials, role-based Snowflake access, and PII redaction of credit-card data.

Mettler-Toledo Mumbai, India	Mar 2024 - Jun 2024
<i>Data Analyst Intern</i>	

- Designed an end-to-end RAG system with fine-tuned T5/BART models (Hugging Face) for automated KPI commentary, with vector search over 10M+ financial records (85%+ relevance); removed 15 hrs/month of manual write-ups.

Marcellus Investment Managers Mumbai, India	Mar 2023 - Nov 2023
<i>Quantitative Analyst Intern</i>	

- Consolidated 25 tables (3M+ rows) across 2 PostgreSQL schemas (ACE Equity, Bloomberg) and created Python automation that cut 10 hrs/week of manual reconciliation.
- Developed a regression-based impact-cost model ($R^2 = 0.82$) for trade-execution optimization and a forensic-screening ML ensemble (84% accuracy, 87% specificity) adopted into ongoing fraud-risk screening.

PROJECTS

- AI Financial Research Agent** · [GitHub] · [Live Demo] · *Full-Stack GenAI, Agentic RAG, MCP, SSE, Evaluation*
- Architected an agentic RAG system over SEC 10-K filings (5 companies, 67K+ chunks) using LangChain, LangGraph ReAct agents, a FastMCP server (streamable-HTTP MCP), ChromaDB with MiniLM embeddings, and Gemini 2.5 Flash Lite; tools discovered at runtime, decoupling agent logic from tool implementations.
 - Built and deployed a full-stack Next.js and TypeScript chat interface streaming source-grounded answers with live citations via Server-Sent Events, over a two-service FastAPI + MCP backend (Docker Compose, MCP-first with direct-agent fallback, 30s timeouts); GitHub Actions CI/CD runs lint, eval, and pytest.
 - Built a RAGAS evaluation harness with a separate judge model to avoid quota contention with the agent's generation model; scored 0.80 faithfulness and 0.80 context recall across question types.